

WEEK 5 PROJECT SUMMARY

HealthConnect Experience Lab — Data Analytics Track

Exploratory Analysis, KPI Development & Business Insights

WEEK 5 EXECUTIVE SNAPSHOT

Week 5 moved the HealthConnect Data Analytics track from analytical planning into practical implementation. The appointment dataset was revalidated, exploratory analysis was completed across the approved business questions, three Week 4 KPI candidates were operationalised, and an initial Power BI dashboard was developed using the decision flow MONITOR → DIAGNOSE → PRIORITIZE. The strongest observed differentiation in No-Show Rates is associated with booking lead time, while recorded previous no-shows provide a useful supporting history signal.

1. WHAT WAS PLANNED FOR WEEK 5

The Week 5 plan was to convert the Week 4 analytical foundation into an evidence-based initial analytics solution. The intended work covered data preparation and quality validation, exploratory analysis of attendance and no-show patterns, development and calculation of selected KPIs, initial dashboard development, interpretation of business findings, formulation of recommendations, documentation of limitations and dependencies, and preparation of the project for the next implementation phase.

2. WHAT WAS COMPLETED

The original HealthConnect Appointment Dataset and Data Dictionary were retained as source resources and were not overwritten. Data types, missing values, duplicate records, categorical values and cross-field consistency were checked. Required type corrections were applied in the analytical workflow, while missing distance-to-clinic and waiting-time values were left unimputed and documented. Exploratory analysis was completed across appointment and scheduling context, reminders, accessibility and operational context, recorded appointment history, and recorded patient characteristics. Three KPIs were operationalised: No-Show Rate, Reminder Coverage Rate, and No-Show Rate by Reminder Status. An initial Power BI dashboard was built around MONITOR → DIAGNOSE → PRIORITIZE, and the principal business insights and recommendations were documented.

3. KEY FINDINGS / ANALYTICAL OUTCOMES

The overall No-Show Rate is 51.15% among 4,737 non-cancelled appointments. Booking lead time provides the clearest observed differentiation: the No-Show Rate rises from 29.47% for appointments booked 0–7 days in advance to 71.36% for those booked 46–60 days in advance. Recorded previous no-shows provide a supporting signal, with the No-Show Rate increasing from 46.30% at zero recorded previous no-shows to 69.74% at three. Reminder Coverage Rate is 72.68%; among non-cancelled appointments, the No-Show Rate is 54.63% without a recorded reminder and 49.86% with a reminder, a descriptive gap of 4.78 percentage points. However, the reminder association is very weak and should not be interpreted causally. Appointment day, appointment time, waiting time, age group and gender did not provide meaningful standalone differentiation; distance to clinic showed only a weak contextual association.

4. MAJOR CHALLENGES ENCOUNTERED

The main analytical challenge was the repeated-patient data quality issue identified in Week 4 and reconfirmed during Week 5. Repeated patient_id values show inconsistencies in recorded demographic and historical attributes, which prevents reliable reconstruction of longitudinal patient histories. A second challenge was distinguishing statistical detectability from decision relevance: several relationships produced statistically detectable results but very small effect sizes. The analysis therefore required disciplined interpretation rather than treating p-values alone as evidence of operational importance. During dashboard construction, Power BI filter interactions were also cross-checked because they could alter displayed values; the affected visuals were corrected and validated against the analytical results.

5. IMPORTANT DECISIONS MADE AND WHY

The appointment record was retained as the unit of analysis because patient-level reconstruction is not sufficiently reliable. Cancelled appointments were excluded from the No-Show Rate denominator but retained in the total appointment population and in Reminder Coverage Rate. The overall No-Show Rate was used as a descriptive reference rather than a benchmark or target. Booking lead-time bands were introduced only after examining the continuous variable, so that the grouped view supported interpretation without replacing the underlying analysis. Recorded previous no-shows were retained as a supporting prioritisation signal rather than a standalone predictor. Reminder exposure was interpreted as an association only; no causal claim about reminder effectiveness was made.

6. CHANGES FROM THE WEEK 4 APPROACH

The Week 4 framework remained valid and was not restarted. Week 5 converted the proposed questions, hypotheses and potential KPIs into actual analysis, calculated measures and visual evidence. The initial KPI candidates were refined into operational definitions with explicit populations and denominators. The analytical approach also became more decision-oriented through an evidence hierarchy: stronger signals were separated from weak or contextual factors, and the dashboard was structured to move from monitoring overall performance to diagnosis and prioritisation.

7. CROSS-TRACK COLLABORATION

A cross-track collaboration was not completed within the Week 5 submission window. Given the remaining time before the submission deadline, priority was placed on completing and validating the Data Analytics track-specific requirements rather than documenting an interaction that had not occurred. This remains an open project dependency for the next phase. A relevant Week 6 collaboration point would be to share the validated Data Analytics findings — particularly booking lead time and recorded previous no-shows — with the Data Science track so that their potential value as modelling features can be assessed. No cross-track output or improvement is claimed for Week 5.

8. REMAINING WORK

The remaining work is to complete final packaging and submission controls: ensure that the notebook, Power BI dashboard, Business Insights & Recommendations document, Week 5 Project Summary and relevant supporting files are correctly named and organised; update the GitHub repository and README where applicable; verify Google Drive access; and complete the submission audit. Analytically, the next phase should test whether combinations of the strongest observed signals improve prioritisation and should further investigate reminder strategy using additional operational context where available.

9. PROPOSED FOCUS FOR WEEK 6

Week 6 should move from initial analytical prioritisation toward refinement and integration. The primary focus should be to test decision-useful combinations of booking lead time and recorded previous no-shows, assess how the analytics can support targeted attendance interventions, and evaluate reminder strategy without assuming causality. Cross-track integration should also begin where relevant, particularly by providing validated analytical findings and KPI definitions to Data Science and documenting how those outputs influence subsequent modelling or solution development. Monitoring logic should remain traceable to the Week 5 evidence base.

WEEK 5 STATUS

Track-specific analytical implementation completed. Final submission packaging, repository/Drive checks and submission-form validation remain to be completed before the deadline.